SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0518**

COURSE OUTCOME COVERED

CO5: *Develop computer vision applications using OpenCV for performing recognition and analysis on images and videos.* (BTL 6)

Assessment Tool Weightage: 10%

Mock Interview Questions

OpenCV Basics & Pipeline Thinking

1. Design and explain a complete OpenCV pipeline to read, process, and display multiple images from a folder efficiently. Clearly justify your design choices and discuss potential challenges and optimizations.

OpenCV Basics & Pipeline Thinking

Pipeline

Input Folder

↓

Read Image

↓

Resize / Grayscale

↓

Noise Removal

↓

Image Processing

↓

Display

↓

Save Result

Pseudocode

```
BEGIN
Get all image files from folder
FOR each image_file DO
   image ← cv2.imread(image_file)
   IF image is NULL THEN
      CONTINUE
   END IF
   resized ← cv2.resize(image, standard_size)
   gray ← cv2.cvtColor(resized, BGR2GRAY)
   filtered ← cv2.GaussianBlur(gray, (5,5), 0)
   result ← cv2.Canny(filtered, 50, 150)
   Display(resized, result)
   Save(result)
END FOR
Close all windows
END
```

Key Points

- **Resize:** reduces processing time.
- **Grayscale:** reduces computational cost.
- **Gaussian Blur:** removes noise.
- **Modular pipeline:** easy to maintain and modify.
- **Challenges:** corrupted images, different resolutions, and large datasets.

- **Optimization:** resize early, process images sequentially, and use parallel processing for large batches.

Image & Video Processing

5. A video stream drops frames during processing. Analyze the problem and redesign your approach to ensure smooth playback, explaining trade-offs and optimization techniques.

Image & Video Processing

Problem

Frames are dropped because **processing time per frame is greater than the video frame interval**.

Redesigned Approach

Video Capture

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Resize Frame

↓

Lightweight Processing

↓

Display

Optimization Techniques

- Reduce frame resolution.
- Convert to grayscale when possible.
- Use efficient algorithms.
- Skip some frames if necessary.
- Use **multithreading**: one thread captures frames while another processes them.

- Avoid unnecessary image copies.

Trade-offs

- **Lower resolution** → faster processing but reduced image quality.
- **Frame skipping** → smoother playback but less temporal information.
- **Lightweight algorithms** → faster response but potentially lower accuracy.
- **Multithreading** → better throughput but increased complexity.

Conclusion: Prioritize real-time performance by reducing processing load and using parallel capture/processing, while accepting small losses in resolution or frame rate when necessary.

Feature Detection & Drawing

11. You need to compare two images and highlight differences visually. Analyze the problem and design a solution, justifying your approach.

Feature Detection & Drawing

Approach

Image 1 + Image 2

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Resize / Align Images

↓

Convert to Grayscale

↓

Absolute Difference

↓

Thresholding

↓

Find Contours

↓

Draw Bounding Boxes

↓

Display Differences

Pseudocode

```
image1 ← cv2.imread("image1.jpg")

image2 ← cv2.imread("image2.jpg")

Resize image2 to image1 size

gray1 ← cv2.cvtColor(image1, GRAY)

gray2 ← cv2.cvtColor(image2, GRAY)

difference ← cv2.absdiff(gray1, gray2)

binary ← cv2.threshold(difference, threshold)

contours ← cv2.findContours(binary)

FOR each contour:

   IF contour area > minimum_area:

      Draw bounding rectangle on image1

Display image1
```

Justification

- **absdiff()** identifies pixel-level changes.

- **Thresholding** removes minor variations/noise.
- **Contours** locate the changed regions.
- **Bounding boxes** make differences easy to visualize.
- Images should be **aligned first**; otherwise, camera movement can appear as false differences.

Face & Expression Recognition

14. A facial expression system fails with slight head movements. Analyze the limitation and propose modifications, justifying your approach.

Face & Expression Recognition

Problem

The system is sensitive to **head pose changes**, causing facial features to shift and reducing expression classification accuracy.

Modifications

Input Image

↓

Face Detection

↓

Face Alignment

↓

Feature Extraction

↓

Expression Classification

Key Improvements

- Use **facial landmarks** to detect eyes, nose, and mouth.
- Apply **face alignment** to normalize rotation and position.
- Use a **pose-tolerant CNN** or train with rotated/angled face images.
- Apply **data augmentation** such as rotation and scaling.

Justification

Face alignment reduces the effect of head movement, while landmark-based normalization provides consistent facial geometry. Training with different poses improves **robustness** to real-world head movements.

Code of Conduct:

I P.V.N. ANISH BABU (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.V.N. ANISH